SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

Dr jaya mabel rani

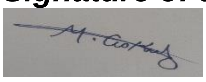
QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	**5**
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	**5**
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	**5**
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	**5**
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	**5**
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	**5**
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	**5**
8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	**5**

9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	**5**
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	**5**

Code of Conduct:

I M. GOKUL 192565067 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high	Shows good	Low	Lacks

		confidence, positive attitude, and professional behavior throughout	confidence with minor issues	confidence; inconsistent behavior	confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

1. Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?

Query processing is the process of converting an SQL query into an efficient method of execution.
Steps:
1. **Parsing and Translation:** The SQL query is checked for syntax and converted into an internal representation.
2. **Query Optimization:** Different execution plans are compared.
3. **Plan Selection:** The optimizer selects the most efficient plan.
4. **Query Execution:** The database executes the selected plan and returns the result.

The **query optimizer** considers factors such as:
- Number of records
- Index availability
- Table size
- Join order
- Disk I/O cost
- CPU and memory usage

The optimizer selects the execution plan with the **lowest estimated cost**.

---

2. What is the difference between heuristic optimization and cost-based optimization? When would you use each?

Heuristic optimization uses predefined rules to improve a query. For example, selections are performed early to reduce the number of records.
Cost-based optimization compares different query execution plans and calculates their estimated cost based on statistics such as table size, indexes, and I/O operations.

Heuristic Optimization	**Cost-Based Optimization**
Uses fixed rules	Uses estimated cost
Faster	More accurate
Less complex	More computationally expensive
Does not require detailed statistics	Requires database statistics

Use heuristic optimization when a quick and simple optimization is needed.
Use cost-based optimization for complex queries where selecting the lowest-cost execution plan is important.

---

3. How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.

In a real project, database connectivity can be implemented using these steps:
1. Install the required database driver.
2. Create a connection between the application and database.
3. Create and execute SQL queries.
4. Process the returned results.

5. Close the connection and release resources.

JDBC (Java Database Connectivity):

- Used mainly with Java applications.
- Provides Java-based database connectivity.
- Platform-independent.

ODBC (Open Database Connectivity):

- A standard interface used by different programming languages.
- Provides connectivity between applications and databases.

Main difference: JDBC is specifically designed for **Java**, while ODBC is a **general database connectivity standard**.

---

4. What are ACID properties? Give a real-world example where each property is critical.

ACID properties ensure reliable and consistent database transactions.

1. **Atomicity:** A transaction is completed fully or not completed at all.
   Example: In a bank transfer, money should not be deducted from one account unless it is added to the other account.
2. **Consistency:** A transaction moves the database from one valid state to another.
   Example: Account balance rules must remain valid after a transaction.
3. **Isolation:** Concurrent transactions should not interfere with each other.
   Example: Two users updating the same bank account should not create incorrect results.
4. **Durability:** Once a transaction is committed, the changes remain permanent.
   Example: A successful online payment should remain recorded even after a system crash.

---

5. Explain dirty read, lost update, and unrepeatable read. How does 2PL solve them?

Dirty Read:
A transaction reads data modified by another transaction before it is committed.
Lost Update:
Two transactions update the same data, and one update overwrites the other.
Unrepeatable Read:
A transaction reads the same data twice and gets different values because another transaction modified it.
Two-Phase Locking (2PL) solves these problems using locks.
It has two phases:

1. **Growing Phase:** A transaction can acquire locks but cannot release them.
2. **Shrinking Phase:** A transaction can release locks but cannot acquire new ones.

By controlling access to shared data, 2PL helps prevent conflicting operations and improves transaction consistency.

---

6. What is deadlock in DBMS? Describe the deadlock detection algorithm and two prevention strategies.

A **deadlock** occurs when two or more transactions wait indefinitely for each other to release resources.
For example:

- T1 holds Resource A and waits for Resource B.
- T2 holds Resource B and waits for Resource A.

Deadlock Detection:

1. Create a **Wait-For Graph**.
2. Each transaction is represented as a node.
3. Draw an edge from T1 to T2 if T1 is waiting for T2.
4. If the graph contains a **cycle**, a deadlock exists.

Prevention strategies:

1. **Resource ordering:** Transactions must request resources in a fixed order.
2. **Wait-Die or Wound-Wait:** Transactions are controlled using timestamps to prevent circular waiting.

Another method is to abort one transaction and release its resources.

---

7. Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?

Immediate Update	Deferred Update
Database may be updated before commit	Database is updated only after commit
Requires Undo and Redo	Mainly requires Redo
More flexible	Simpler recovery
Uncommitted changes may need to be reversed	Uncommitted changes are not written to the database

Immediate Update is preferred in systems where transactions need to update data immediately and efficient rollback is required.
Deferred Update is preferred when simplicity and safety are important because uncommitted changes are not directly written to the database.

---

8. Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?

Shadow Paging is a recovery technique where the database maintains:

- A **shadow page table** representing the old stable database.
- A **current page table** representing updated data.

During a transaction, modified pages are written to new locations. After a successful commit, the current page table becomes the new database state.

Advantages:

- Does not require traditional Undo operations.
- Recovery is simple.
- Provides quick crash recovery.
- Avoids complex log processing.

Limitations:

- Requires additional storage.
- Page tables may need frequent copying.
- Not suitable for high-concurrency systems.
- Can cause storage fragmentation.

Compared with shadow paging, **log-based recovery** is generally more suitable for large and highly concurrent databases.

---

9. What is a serializability schedule? How do you test conflict serializability using a precedence graph?

A **serializable schedule** is a concurrent execution of transactions that produces the same result as some serial execution of those transactions.

To test **conflict serializability**:

1. Create a node for each transaction.
2. Identify conflicting operations on the same data item.
3. Add an edge from Ti to Tj if Ti's operation occurs before and conflicts with Tj's operation.
4. Construct the **precedence graph**.
5. Check for a cycle.

Result:

- **No cycle:** The schedule is conflict serializable.
- **Cycle exists:** The schedule is not conflict serializable.

A topological order of an acyclic graph gives the equivalent serial order.

---

10. Describe Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?

Basic 2PL:

- Has a growing phase and shrinking phase.
- Locks are acquired first and later released.
- Ensures conflict serializability.
- May still cause cascading rollback.

Strict 2PL:

- Exclusive/write locks are held until the transaction commits or aborts.
- Prevents dirty reads and cascading rollback.
- Provides better recoverability.

Rigorous 2PL:

- Both shared/read and exclusive/write locks are held until commit or abort.
- Provides strong consistency and strict transaction ordering.